MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 11** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | **(6 × 1 = 6 Marks)**

Q1. Quadrilateral $ABCD$ circumscribes a circle. If $AB = 9$ cm, $BC = 10$ cm, and $AD = 7$ cm, then CD equals: [1]

(A) 8 cm (B) 7 cm (C) 6 cm (D) 9 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) 8 cm
Justification: $AB + CD = BC + AD \implies 9 + CD = 10 + 7 \implies CD = 17 - 9 = \boxed{\mathbf{8}\text{ cm}}$.

Q2. If two tangents PA and PB are drawn from point P to a circle of radius 5 cm with centre O such that $OP = 13$ cm, then area of quadrilateral $PAOB$ is: [1]

(A) 30 cm^2 (B) 60 cm^2 (C) 65 cm^2 (D) 120 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 60 cm^2
Justification: Tangent $PA = \sqrt{13^2 - 5^2} = 12$ cm. Area$(PAOB) = 2 \times \frac{1}{2}(5)(12) = \boxed{\mathbf{60}\ \text{cm}^2}$.

Q3. If the length of a tangent from a point at distance 5 cm from the centre of a circle is 4 cm, then the diameter of the circle is: [1]

(A) 3 cm (B) 6 cm (C) 8 cm (D) 10 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 6 cm
Justification: Radius $r = \sqrt{5^2 - 4^2} = 3$ cm $\implies$ Diameter $= 2r = \boxed{\mathbf{6}\text{ cm}}$.

Q4. Two concentric circles are of radii 20 cm and 12 cm. The length of the chord of the larger circle tangent to the smaller circle is: [1]

(A) 16 cm (B) 24 cm (C) 32 cm (D) 36 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 32 cm

Justification: Half-chord $= \sqrt{20^2 - 12^2} = \sqrt{256} = 16$ cm $\implies$ Chord $= 2 \times 16 =$ **32** cm.

Q5. Tangents drawn at the endpoints of a diameter of a circle are always: [1]

(A) Perpendicular (B) Intersecting (C) Parallel (D) Coincident

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) Parallel

Justification: Both tangents make $90°$ angles with the diameter, so alternate interior angles are equal ($90°$), proving they are **Parallel**.

Q6. **Assertion (A):** If the distance between two parallel tangents to a circle is 12 cm, then the radius of the circle is 6 cm.
Reason (R): The line segment joining the points of contact of two parallel tangents is a diameter of the circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: By (R), the distance between two parallel tangents equals the diameter, so $2r = 12$ cm $\implies r = 6$ cm. Both statements are true and (R) explains (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | ($2 \times 2 = 4$ Marks)

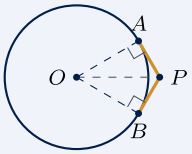
Q7. In a circle with centre O, two tangents PA and PB are drawn such that $\angle APB = 120°$. Prove that $OP = 2AP$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Angle Bisection via Congruence [1 Mark]
$\triangle OAP \cong \triangle OBP$ (RHS), so OP bisects $\angle APB$: $\angle OPA = \frac{1}{2}(120°) = 60°$.

Step 2: Trigonometric Ratio in $\triangle OAP$ [1 Mark]
Since $\angle OAP = 90°$, $\cos 60° = \frac{AP}{OP} \implies \frac{1}{2} = \frac{AP}{OP}$.

$\therefore OP = 2AP$ [Proved]



Q8. Two concentric circles are of radii 25 cm and 20 cm. Find the length of the chord of the larger circle which touches the smaller circle. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

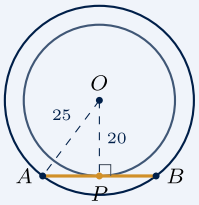
Let O be common centre and AB touch the inner circle at P ($OP \perp AB$, $AP = PB$).

Step 1: Pythagoras Theorem in $\triangle OPA$ [1 Mark]

$AP = \sqrt{OA^2 - OP^2} = \sqrt{25^2 - 20^2} = \sqrt{225} = 15$ cm.

Step 2: Chord Length [1 Mark]

$$\boxed{AB = 2 \times AP = \mathbf{30}\text{ cm}}$$



SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. If $\triangle ABC$ circumscribes a circle with radius r, prove that $\text{Area}(\triangle ABC) = r \cdot s$, where s is the semi-perimeter. If area is 84 cm^2 and perimeter is 28 cm, find the inradius r. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

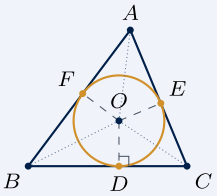
Step 1: Proving Formula $\text{Area}(\triangle ABC) = r \cdot s$ [2 Marks]

Let circle touch BC, CA, AB at D, E, F ($OD = OE = OF = r$ perpendicular to sides).

$\text{Area}(\triangle ABC) = \text{Area}(\triangle OBC) + \text{Area}(\triangle OCA) + \text{Area}(\triangle OAB) = \frac{1}{2}BC \cdot r + \frac{1}{2}CA \cdot r + \frac{1}{2}AB \cdot r = r\left(\frac{a+b+c}{2}\right) = r \cdot s.$ [Proved]

Step 2: Computing Inradius r [1 Mark]

$s = \frac{28}{2} = 14$ cm $\implies 84 = r \times 14 \implies \boxed{r = \mathbf{6}\text{ cm}}$.



Q10. Two tangents PA and PB are drawn to a circle with centre O such that $\angle APB = 60°$. A third tangent is drawn touching the circle at C and intersecting PA, PB at D, E. Find $\angle DOE$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Finding Central Angle $\angle AOB$ [1 Mark]

Since $\angle OAP = \angle OBP = 90°$, $\angle AOB = 180° - \angle APB = 180° - 60° = 120°$.

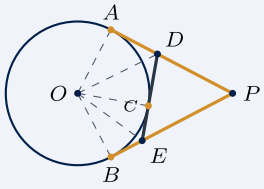
Step 2: Halving Central Angle [2 Marks]

In $\triangle OAD$ and $\triangle OCD$: $OA = OC$, $OD = OD$, $DA = DC \implies \triangle OAD \cong \triangle OCD$ (SSS) $\implies \angle AOD = \angle COD$.

Similarly $\angle BOE = \angle COE$.

$\angle AOB = 2\angle COD + 2\angle COE = 2\angle DOE \implies$

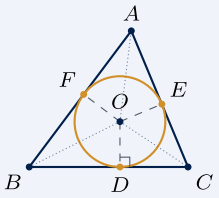
$$\boxed{\angle DOE = \frac{120°}{2} = \mathbf{60°}}.$$



Q11. Prove that the lengths of tangents drawn from an external point to a circle are equal. Using this theorem, if $\triangle ABC$ is circumscribed about a circle of radius r touching sides BC, CA, AB at D, E, F, prove that: (i) $AF + BD + CE = AE + BF + CD = \frac{1}{2}$(Perimeter of $\triangle ABC$), (ii) Area$(\triangle ABC) = r \cdot s$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Theorem 10.2 Proof: [2 Marks]
Given tangents TP, TQ from external point T to circle (O). Since $\angle OPT = \angle OQT = 90°$, $OT = OT$, $OP = OQ$, by RHS $\triangle OPT \cong \triangle OQT \implies TP = TQ$ (CPCT).



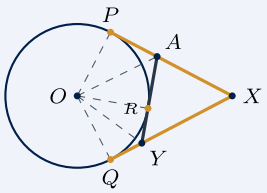
(i) Equal Half-Perimeter Tangent Sums: [1.5 Marks]
By Theorem 10.2: $AF = AE$, $BD = BF$, $CE = CD$.
Perimeter $= (AF + BF) + (BD + CD) + (CE + AE) = 2(AF + BD + CE) = 2(AE + BF + CD)$.
Thus $AF + BD + CE = AE + BF + CD = \frac{1}{2}$Perimeter$(\triangle ABC)$. [Proved]

(ii) Proving Area$(\triangle ABC) = r \cdot s$: [1.5 Marks]
Joining OA, OB, OC: Area$(\triangle ABC) = \frac{1}{2}BC \cdot r + \frac{1}{2}CA \cdot r + \frac{1}{2}AB \cdot r = r \cdot \frac{AB+BC+CA}{2} =$ $\boxed{r \cdot s \quad \text{[Proved]}}$.

Q12. In the given figure, XP and XQ are two tangents to a circle with centre O from a point X outside the circle. ARY is a tangent to the circle at a point R intersecting XP and XQ at A and Y respectively. If $XP = 15$ cm, find the perimeter of $\triangle XAY$. If $\angle PXQ = 50°$, find $\angle AOY$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Part 1: Finding Perimeter of $\triangle XAY$ [2.5 Marks]
By Theorem 10.2: $XP = XQ = 15$ cm, $AP = AR$, and $YQ = YR$.
Perimeter$(\triangle XAY) = XA + AY + YX = XA + (AR + RY) + YX = (XA + AP) + (YQ + YX) = XP + XQ = 15 + 15 =$ **30** cm.



Part 2: Finding Angle $\angle AOY$ [2.5 Marks]
In quad $OPXQ$, $\angle OPX = \angle OQX = 90° \implies \angle POQ = 180° - 50° = 130°$.
Join OP, OR, OQ. Since $\triangle OAP \cong \triangle OAR$ and $\triangle OYQ \cong \triangle OYR$ (SSS): $\angle AOP = \angle AOR$ and $\angle YOQ = \angle YOR$.
Thus $\angle POQ = 2(\angle AOR + \angle YOR) = 2\angle AOY \implies \angle AOY = \frac{130°}{2} = 65°$.

$$\boxed{\text{Perimeter}(\triangle XAY) = \mathbf{30} \text{ cm}, \quad \angle AOY = \mathbf{65°}}$$

Q13. Gear Transmission Meshing between Tangent Pulleys

Two machine pulleys of radii $r_1 = 9$ cm and $r_2 = 4$ cm touch each other externally at point C. A direct common drive belt AB touches the first pulley at A and the second at B. A common tangent at the contact point C intersects the drive belt AB at point P.

(i) State the relation between segments PA, PB, and PC. [1]

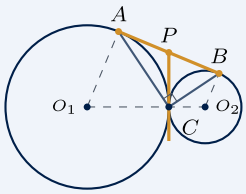
(ii) Find the measure of angle $\angle ACB$. [1]

(iii) Calculate the total length of common tangent segment AB, OR calculate the distance between the pulley centres O_1O_2. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Relation Between PA, PB, PC: [1 Mark]

By Theorem 10.2 from P: $PA = PC$ and $PB = PC$, so $\boxed{\mathbf{PA = PB = PC}}$ (P bisects AB).



(ii) Measure of $\angle ACB$: [1 Mark]

$\angle PCA = \angle PAC$ and $\angle PCB = \angle PBC$. In $\triangle ABC$, sum gives $2(\angle PCA + \angle PCB) = 180^\circ \implies \boxed{\mathbf{\angle ACB = 90^\circ}}$.

(iii) Main Choice — Length of Direct Common Tangent AB: [2 Marks]

Centre distance $O_1O_2 = r_1 + r_2 = 9 + 4 = 13$ cm, and perpendicular height difference $r_1 - r_2 = 9 - 4 = 5$ cm.

Thus $AB = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = \sqrt{13^2 - 5^2} = \sqrt{144} = \boxed{\mathbf{12}\text{ cm}}$.

— OR — (Alternative Choice — Distance Between Centres O_1O_2): [2 Marks]

Since the pulleys touch externally at C, $O_1O_2 = r_1 + r_2 = 9 + 4 = \boxed{\mathbf{13}\text{ cm}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★